

# Han-Teng Liao

Socio-Technical System Architect • Ex Associate Professor and Vice Dean

Taipei and Penang

✉ Email 🏠 Project 🆔 0000-0003-1081-5599 📄 Google Scholar 🌐 Web of Science 🐙 GitHub 🔗 LinkedIn 🐦 twitter

## Professional Service

### Peer Reviewers

*Editorial*

Completed 25+ reviews (mostly Q1 and Q2 journals), esp. on AI and Industry 5.0.

2021 — Present

- **Scientific Reports** (Q1:2) **Financial Innovation** (Q1:3); **Nature Humanities and Social Sciences Communications** (Q1:1)
- **Politics and Governance** (Q2:1), **Frontiers in Public Health** (Q1:1); **Journal of Clinical Epidemiology** (Q1:1)
- **Heliyon** (Q1:2); **Cogent Economics & Finance** (Q1:1); **Cogent Education** (Q2:1); **Applied Economics Letters** (Q3:1)
- **Future Internet** (Q2:1); **Discover Artificial Intelligence** (1); **Discover Education** (1)
- **Sustainability** (Q2:4); **Environment Systems and Decisions** (1)

### Technical Committee Member

*Technical*

**Joint IEEE Systems Engineering/TEMS TC on Carbon Neutrality**

2026 — Present

- Explore active research and strategies to promote carbon neutrality best practices, driving consortium actions with bodies like **SiNANO** and orchestrating strategic liaisons across IEEE Technical Committees and international organizations.

### Expert Contributor

*International Telecommunication Union*

ITU-T Focus Group on AI for Agriculture (FG-AI4A)

2023 - 2024

- Authored technical input document b-FG-AI4A-I-132 on linking carbon accounting data with agricultural business.
- Successfully integrated the framework into ITU-T Y.Suppl.83 (07/2024) under Section 7.2.7 (TG-Digital Agri-Finance).
- Translates 'Carbon Information Chains' regional best practices into international standard reference.

### Student Branch Development & Launch

*Guangzhou, China*

IEEE Technology and Engineering Management Society (TEMS)

2019 - 2023

- Orchestrated and managed the establishment of the IEEE TEMS Student Branch during tenure as Center Director.

### Program Committee Chair

*Guangzhou, China*

Management Science Informatization and Economic Innovation Development Conference (MSIEID)

December 18-20, 2020

### Conference Special Session Chair

*Guangzhou, China*

World Eco-Design Conference (WEDC) 2019

2019

- Chaired the session on **Internet Eco-Design: Sustainable Development Innovation Design and Internet Eco-systems** (第二届世界生态设计大会-互联网生态设计：可持续发展创新设计与互联网生态).

### Program Committee Member

*Yogyakarta, Indonesia*

14th International Conference of IFIP Working Group 9.4

2017

### Invited Expert

*United Nations ESCAP*

Expert Meeting on Big Data and the 2030 Agenda for Sustainable Development

2015

- Provided expert insights on leveraging big data to advance the UN 2030 Sustainable Development Goals (SDGs).

### Visiting Expert

*Leiden University, Netherlands*

ERC Project "Authoritarianism 2.0"

2014

- Invited by Professor Daniela Stockmann to advise the European Research Council (ERC) project team on digital methods.

### Conference Organizer

*Washington DC, US*

Chinese Internet Research Conference (CIRC 2011)

2011

- Co-organized the conference theme "Chinese Factors in Internet Development: Global Public Goods, National Policies, and Private Interests."

## Education

---

### DPhil (Ph.D.) in Information, Communication, and Social Sciences

University of Oxford

Oxford, United Kingdom

10/2007 - 10/2014

### M.Sc. in Computer Science and Information Engineering

National Taiwan University

Taipei, Taiwan

01/08/2000 - 01/06/2004

### M.S.S. (Master of Social Science) in Journalism

National Taiwan University

Taipei, Taiwan

01/08/2000 - 01/06/2004

### B.Sc. in Electrical Engineering

National Taiwan University (Includes a year-long exchange at the University of British Columbia)

Taipei, Taiwan

01/10/1994 - 01/06/2000

### Bachelor of Arts in Foreign Languages and Literatures

National Taiwan University

Taipei, Taiwan

01/08/1995 - 01/06/2000

## Work Experience

---

### Editor & Project Founder (Public Service Initiative)

scvcs.footprints.cv - SSbD Chips Value Chain Services | GBS.today

Penang, Malaysia

01/01/2026 - Current

### Data Scientist and CEO

Oxford Roadmapping

Guangzhou, China

01/01/2023 - Current

### Director, Higher Education Impact Assessment Center

Nanfeng College of Sun Yat-Sen University

Guangzhou, China

01/12/2019 - 01/01/2023

### Associate Professor & Vice-Dean, Internet and New Media

Nanfeng College of Sun Yat-Sen University

Guangzhou, China

01/01/2017 - 01/01/2021

### Research Fellow

United Nations University (UNU) Institute on Computing and Society

Macau, China

01/07/2015 - 01/11/2016

### Yahoo! Fellow, Institute for the Study of Diplomacy

Georgetown University

Washington DC, United States

01/08/2010 - 01/07/2011

### Doctoral Research Fellow

Institute of Sociology, Academia Sinica

Taipei, Taiwan

01/08/2009 - 01/06/2010

### Political Warfare Officer (Mandatory Service)

Taiwan Military

Taiwan

01/11/2005 - 01/12/2006

### Project Manager

Open Source Software Foundry (OSSF), Institute of Information Science, Academia Sinica

Taipei, Taiwan

01/10/2004 - 30/11/2005

### Action Researcher & Graduate Teaching Assistant

Institute of Journalism, National Taiwan University

Taipei, Taiwan

01/09/2002 - 01/12/2003

### Programmer & Independent Developer (Digital Publishing)

Yuan-Liou Publishing Co., Ltd.

Taipei, Taiwan

01/07/2000 - 31/08/2000

## Awards

---

### Certificate of Merit

Recognized for outstanding academic contributions

IEEE TEMSMET Conference

2021

- Awarded for CSR (Corporate Social Responsibility) and ESG (Environmental, Social, and Governance) reporting research.

## Province-Level ‘First Class’ Course

Curriculum Design & Pedagogy Excellence

Guangdong Department of Education

2020

- Recognized for the development and instruction of the Capstone Project: ‘API, machine learning and AI’.

## Nation-Level First Prize

Instructional Excellence

National College Art & Design Awards

2020

- Awarded for Best Teaching for Capstone Project: ‘API, machine learning and AI’.

## OxTALENT Awards

Best Interactive Infographics

University of Oxford

2014

- Recognized excellence in university teaching using interactive digital content.

## Best Student Paper Award

Research Excellence

Chinese Internet Research Conference

2009

- Awarded for foundational digital community research.

## PGP Corporation and OII Scholarship

Merit-based Doctoral Scholarship

Oxford Internet Institute, University of Oxford

2007

- Value: £12,000 to support DPhil research.

## Taiwan Merit Scholarship

Doctoral Research Fellowship

Taiwan Government

2007

- Awarded for doctoral research in Digital Content (TWD \$2,000,000).

## Publications

---

### From Stacks to Circuits: A Regenerative Socio-Technical Roadmap for AI Infrastructure within Planetary Boundaries

ICE IEEE/ITMC, 22nd-25th June 2026, Porto - Portugal

Conference Paper

June 2026

- Contributors: **Han-Teng Liao**; Karen Ang
- Challenges Nvidia’s “electrons-to-tokens” 5-layer scaling trajectories by converting thermodynamic entropy and Scope 3 metrics into dynamic, regeneratively accountable circuits.
- Introduces a Regenerative Socio-Technical (RST) Roadmap to address the hidden environmental, thermal, and material costs of unconstrained Generative AI scaling.
- (To be) Presented in **Digital Circular Economy** Special Session, providing the conceptual foundation for the **SCVCS** to anchor its **AuditSoS** Pillar through metrology-grounded Digital Product Passports (DPP).

### Scoping Review of AI, Metrology, and ESG in the Semiconductor Sector: Implications for Safe and Sustainable by Design (SSbD)

ICE IEEE/ITMC, 22nd-25th June 2026, Porto - Portugal

Conference Paper

June 2026

- Contributors: Karen Ang; **Han-Teng Liao**
- Proposes a novel 6-layer System of Systems (SoS) architecture to integrate AI-driven metrology, supply chain ESG metrics, and federated industrial data spaces into an integrated architecture.
- Bridges the gap between precision manufacturing and verifiable environmental sustainability by embedding SSbD as core operational logic for Industry 5.0.
- (To be) Presented in **Technology** Session, providing the conceptual foundation for the **SCVCS** to integrate RealSoS, GuardSoS and AuditSoS Pillars as a System of Systems(SoS).

### Orchestrating the Twin Transition in Multinational Corporations: Technology Roadmapping for Green and Digital Global Business Services

ICE IEEE/ITMC, 22nd-25th June 2026, Porto - Portugal

Conference Paper

June 2026

- Contributors: **Han-Teng Liao**; Karen Ang

- Synthesizes Technology Roadmapping (TRM) and the ITU innovation toolkit to treat Global Business Services (GBS) units as strategic “operational airlocks.”
- Establishes a socio-technical framework for multinational corporations to harmonize internal digital agility and AI-native transitions with external sustainability mandates.
- (To be) Presented in **Innovation** Session, providing the conceptual foundation for the **SCVCS** to cross-map carbon border adjustment data aggregation under its **AuditSoS** Pillar for regional GBS 5.0 networks.

## **Collaborating on ESG consulting, reporting, and communicating education: Using partner maps for capability building design**

*Frontiers in Environmental Science*

Journal article

March 2023

- Contributors: **Han-Teng Liao**; Chung-Lien Pan; Yuan Zhang
- Applied design science and circular economy frameworks to map the specific data and digital skills required for ESG consulting and reporting.
- Establishes a collaborative roadmap for universities and industry to build capability for sustainable innovation across disciplines and sectors.

## **Knowledge Mapping of Resilience and Human Rights in Supply Chains**

*Frontiers in Environmental Science*

Journal article

April 2023

- Contributors: Chung-Lien Pan; **Han-Teng Liao** (Corresponding Author); Yuan Zhang
- Developed a roadmapping taxonomy to integrate human rights and resilience into supply chain architecture.
- Provides a governance framework for the Twin Transition, offering strategic tools for risk mitigation and value-based alignment in complex, cross-border industrial ecosystems.

## **Knowledge Mapping Analysis of Intelligent Ports: Research Facing Global Value Chain Challenges**

*Systems*

Journal article

2023

- Contributors: **Han-Teng Liao**; Tsung-Ming Lo; Chung-Lien Pan
- Utilized scientometrics and science mapping to analyze the digital transformation of intelligent ports within Global Value Chains (GVC).
- Identifies technical and governance bottlenecks in open and collaborative innovation, providing a strategic roadmap for port-city digital integration.

## **Carbon neutrality policies and technologies: A scientometric analysis of social science disciplines'**

*Frontiers in Environmental Science*

Journal article

October 2021

- Contributors: Yuan Zhang; Chung-Lien Pan; **Han-Teng Liao**
- Conducted a comprehensive science mapping and systematic review of 907 social science articles to delineate disciplinary boundaries and collaborative frameworks for net-zero pathways. - Highlighted the critical intersection where green digital transformation and datafied environmental indicators must guide human production and consumption patterns.

## **A Literature Review of Museum and Heritage on Digitization, Digitalization, and Digital Transformation'**

*Proceedings of the 6th International Conference on Humanities and Social Science Research (ICHSSR 2020)*

Conference paper

May 2020

- Contributors: **Han-Teng Liao**; Man Zhao; Si-Pan Sun
- Executed a scientometric analysis of 1,107 Web of Science articles using VOSviewer and custom Python scripts to categorize distinct layers of socio-technical transitions. - Mapped the shift in cultural institution research fronts from technology-centric concerns to people-centric concerns (digital literacy, organizational adaptation).

## **Conflict and consensus in the Chinese version of Wikipedia**

*IEEE Technology and Society Magazine*

Journal article

June 2009

- Contributors: **Han-Teng Liao**
- Investigated the socio-technical architectures, editorial policy alignments, and hyperlinked structures used to govern consensus in high-conflict, cross-border digital environments. - Analyzed how political and cultural cybernetics operate when collaborative web spaces cross sensitive jurisdictional and ideological boundaries.

More publications and citation indices are available via my **ORCID Profile** and **Google Scholar Profile**.

*Academic Profiles*

Impact Information

*Current*

- Includes comprehensive tracking of peer-reviewed articles, conference papers, and h-index metrics.

## Skills

---

**Strategic Architecture & Governance**

**Intelligence & Foresight (Data Science)**

**AI, Data Architecture & Automation**

**Product & Agile Leadership**

**Language Skills**